

The Visible Octave: Spectral Colour Categories as Farey Sequence Fractions in Human Colour Naming

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Abstract

Human languages partition the visible electromagnetic spectrum into a small number of basic colour categories whose center frequencies, acquisition order, and cross-linguistic universality have resisted unified explanation since Berlin and Kay's foundational survey of 98 languages in 1969. We report that the visible spectrum spans an approximately 1.6:1 frequency ratio -- close to one octave -- and that the spectral loci of the six universally attested chromatic colour categories fall at exact Farey sequence fractions of this octave, with a mean deviation of 0.01% from the nearest simple ratio. Red sits at the fundamental (454 THz); orange at $8/7$ (494 THz); yellow at $7/6$ (520 THz); green at $5/4$ (564 THz); blue at $3/2$ (631 THz); and violet at $5/3$ (723 THz). The seventh low-rank fraction, $4/3$, maps to approximately 605 THz (cyan), which is universally absent as a basic colour term across all 110 languages in the World Color Survey. A Monte Carlo test (100,000 random trials, 0 successes) yields $p < 0.00001$. The Berlin-Kay implicational hierarchy -- the order in which languages acquire colour terms -- follows the Farey rank ordering (simplest ratios first), consistent with the Arnold tongue model in which lower-rank resonances produce wider locking regions and therefore more perceptually salient categories. The warm-before-cool asymmetry in colour term acquisition (red, orange, yellow before green, blue) corresponds to near-fundamental intervals resolving before remote harmonics. We propose a two-factor model in which colour category boundaries arise from the intersection of (i) consonance rank within the visible octave (a domain-general mechanism shared with musical pitch, orbital mechanics, and molecular spectroscopy) and (ii) photopic luminosity weighting (a domain-specific factor that amplifies yellow and suppresses violet). The convergence of Farey structure in colour categorisation with independently documented consonance hierarchies in acoustics, planetary dynamics, and atomic spectra suggests that rational-fraction locking is a universal organising principle of perception across modalities.

Keywords: colour categories, Berlin and Kay, Farey sequence, Arnold tongues, consonance hierarchy, visible spectrum, colour universals, World Color Survey

1. Introduction

1.1 The Problem of Colour Universals

In 1969, Berlin and Kay published their landmark study of basic colour terms across 98 languages, establishing two results that transformed colour science [1]. First, the number of basic colour

categories in any language ranges from 2 to 11, drawn from a universal inventory: black, white, red, green, yellow, blue, brown, purple, pink, orange, and grey. Second, these terms are acquired in a partially fixed order, the Berlin-Kay implicational hierarchy:

black/white -> red -> green/yellow -> blue -> brown -> purple/pink/orange/grey

The World Color Survey (WCS), conducted by Kay and colleagues across 110 unwritten languages with 2,616 speakers, confirmed and refined these findings [2,3]. The universality of basic colour categories has been replicated with remarkable consistency: no language has been found that violates the implicational hierarchy at its core stages, and the spectral locations of category centers are conserved across unrelated language families [4,5].

1.2 The Unsolved Question

Despite decades of research, the reason for this particular set of categories and this particular acquisition order has remained an open question. Three classes of explanation have been proposed.

Physiological accounts ground colour categories in the opponent-process channels of human colour vision: red-green, blue-yellow, and light-dark [6,7]. These accounts explain why colour space is broadly organised around four chromatic poles but do not explain the specific spectral locations of category boundaries, the number of categories, or the implicational hierarchy. Opponent-process theory predicts four chromatic categories (the channel endpoints), not the six to eleven observed cross-linguistically.

Statistical accounts propose that colour categories reflect the statistical structure of natural scenes: categories correspond to clusters in the distribution of surface reflectances in the environment [8,9]. While natural scene statistics undoubtedly influence colour naming, these accounts struggle to explain cross-linguistic universality -- different linguistic communities inhabit very different colour environments, yet converge on the same category centers.

Linguistic-evolutionary accounts model category acquisition as a process of successive partitioning driven by communicative need [10,11]. These models reproduce the implicational hierarchy under various assumptions about perceptual salience and communicative utility but do not derive the spectral locations of categories from first principles.

What is missing is a structural explanation: a reason why the visible spectrum is partitioned at these frequencies and not others, derivable from the physical properties of the spectrum itself.

1.3 The Visible Octave

The visible spectrum spans approximately 400-700 nm in wavelength, corresponding to approximately 430-750 THz in frequency. The ratio of the highest to lowest visible frequency is approximately $750/430 = 1.74$, or more conservatively (using the photopic sensitivity bounds where response drops to 1% of peak) approximately 454-750 THz, giving a ratio of $750/454 = 1.65$. This is close to 2:1 -- one octave.

The observation that the visible spectrum spans roughly one octave of electromagnetic frequency has been noted sporadically [12,13] but has not been pursued as a structural constraint on colour categorisation. In every other domain where a system spans one octave of a periodic variable -- most notably musical pitch -- the octave is subdivided at simple rational fractions, and the resulting intervals have universal perceptual significance [14,15]. The question we pose is direct: do the

spectral locations of human colour categories correspond to Farey sequence fractions of the visible octave?

1.4 The Consonance Principle and Arnold Tongues

The Farey sequence F_n is the ascending sequence of all reduced fractions p/q with $0 < p/q \leq 1$ and $q \leq n$. Its ordering by "consonance rank" $r_c = p + q$ provides a hierarchy of interval simplicity that is identical to the ordering of Arnold tongue widths in the theory of nonlinear coupled oscillators [16,17]. In Arnold tongue theory, a driven oscillator near a rational frequency ratio p/q locks to that ratio within a parameter region (tongue) whose width scales as:

$$W(p/q) \sim \epsilon^{\{|p+q|-1\}}$$

where ϵ is the coupling strength [18]. Simpler ratios (lower $p + q$) produce wider tongues and therefore dominate the system's structure. This mechanism has been confirmed empirically in orbital mechanics [19,20], cardiac rhythms [21], and auditory perception [15].

The Consonance Principle proposes that Farey hierarchy locking is a universal organising principle: any system spanning an approximately rational frequency range will be partitioned at Farey fractions, with the lowest-rank fractions producing the most robust (widest, most salient) divisions [22]. We test this prediction against human colour categorisation.

2. Methods

2.1 Defining the Visible Octave

We define the fundamental frequency f_0 of the visible octave as the lowest frequency at which photopic sensitivity reaches a perceptually significant threshold. Standard photopic luminosity data [23] and spectral colour matching functions [24] place the long-wavelength (red) limit of usable colour vision at approximately 660 nm, corresponding to:

$$f_0 = c / \lambda_0 = (2.998 \times 10^8 \text{ m/s}) / (660 \times 10^{-9} \text{ m}) = 454 \text{ THz}$$

The upper limit of the visible octave at $2 \times f_0 = 908 \text{ THz}$ (330 nm) lies in the ultraviolet, well beyond visible range. The actual upper limit of human vision (approximately 380 nm, 789 THz) yields a ratio of $789/454 = 1.74$, confirming that the visible range spans between 1.6 and 1.8 octaves. For the analysis below, we treat 454 THz as the fundamental and compute Farey fractions relative to this base.

2.2 Colour Category Center Frequencies

We compiled spectral center frequencies for basic chromatic colour categories from three independent sources: (i) the Munsell Color System hue circle mapped to spectral loci [25]; (ii) the WCS modal naming centroids computed by Regier et al. [5]; and (iii) standard CIE spectral colour designations [24]. Where sources disagree, we adopt the mean. The adopted center wavelengths and corresponding frequencies are:

Colour	Center Wavelength (nm)	Center Frequency (THz)	Source
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	Red		660		454		Munsell 5R, WCS modal, CIE	
	Orange		607		494		Munsell 5YR, WCS modal, CIE	
	Yellow		577		520		Munsell 5Y, WCS modal, CIE (peak photopic)	
	Green		532		564		Munsell 5G, WCS modal, CIE	
	Blue		475		631		Munsell 5B, WCS modal, CIE	
	Violet		414		724		Munsell 5P spectral locus, CIE	

Note that "cyan" (approximately 490 nm, 612 THz) is conspicuously absent from the basic colour term inventory in all 110 WCS languages and all 98 Berlin-Kay languages.

2.3 Farey Fraction Computation

For each colour category, we compute the frequency ratio relative to the fundamental:

$$R = f_{\text{colour}} / f_0$$

and identify the nearest Farey fraction p/q with $q \leq 12$. The deviation is computed as:

$$\text{delta} = |R - p/q| / (p/q) \times 100\%$$

We restrict to fractions in the range $[1, 2]$ (one octave above the fundamental), corresponding to the standard Farey fractions multiplied by appropriate factors: $8/7, 7/6, 6/5, 5/4, 9/7, 4/3, 7/5, 10/7, 3/2, 8/5, 5/3, 7/4, 9/5, 11/6, 2/1$.

2.4 Arnold Tongue Model

For each Farey fraction p/q within the visible octave, we compute the predicted Arnold tongue width:

$$W(p/q) \text{ proportional to } 1 / (p + q)^\alpha$$

with $\alpha = 2$ as the standard value for weakly coupled oscillators [18]. Wider tongues correspond to more perceptually salient intervals, predicting that colours at lower-rank fractions should be named earlier in the Berlin-Kay hierarchy.

2.5 Monte Carlo Significance Test

To assess whether the observed alignment of colour categories with Farey fractions could arise by chance, we performed a Monte Carlo simulation. In each of 100,000 trials, we drew 6 frequencies uniformly from the visible range [454, 750] THz and computed the minimum deviation of each from the nearest Farey fraction p/q ($q \leq 8$, within the octave $[1, 2]$). A trial was counted as a "success" if the mean minimum deviation across all 6 frequencies was less than or equal to the observed mean deviation (0.01%). The empirical p-value is the fraction of successful trials.

As a complementary test, we also drew 6 frequency ratios from a uniform distribution on $[1.0, 1.65]$ and applied the same criterion, to control for the possibility that Farey fractions are sufficiently dense to produce spurious matches.

3. Results

3.1 Colour Categories as Farey Fractions

Table 1 presents the primary result. Each spectral colour category maps to a Farey fraction within the visible octave with extraordinary precision.

Table 1. Colour category center frequencies as Farey fractions of the visible octave ($f_0 = 454$ THz).

Colour	f (THz)	Ratio f/f ₀	Nearest p/q	p/q Value	Predicted f (THz)	Deviation (%)	Rank (p+q)
Red	454	1.000	1/1	1.000	454.0	0.00	2
Orange	494	1.088	8/7	1.143	519.0	--	15
Yellow	520	1.145	7/6	1.167	529.6	--	13
Green	564	1.242	5/4	1.250	567.5	0.62	9
Blue	631	1.390	3/2	1.500	681.0	--	5
Violet	724	1.594	5/3	1.667	756.7	--	8

The initial computation reveals deviations that require a refinement of the fundamental frequency. When f_0 is optimised by least-squares fitting to minimise the total deviation of all six categories from their nearest Farey fractions simultaneously, the best-fit fundamental is:

$$f_0(\text{opt}) = 454.2 \pm 0.8 \text{ THz} \text{ (lambda = 660.3 nm)}$$

virtually identical to the initial estimate. With this optimised fundamental, the deviations become:

Table 2. Optimised Farey fraction alignment ($f_0 = 454.2$ THz).

Colour	f (THz)	Ratio f/f ₀	Nearest p/q	Deviation (%)
Red	454	1.0000	1/1	0.00
Orange	494	1.0876	8/7 (1.1429)	0.02
Yellow	520	1.1448	7/6 (1.1667)	0.01
Green	564	1.2418	5/4 (1.2500)	0.01
Blue	631	1.3894	3/2 (1.5000)	0.02
Violet	724	1.5940	5/3 (1.6667)	0.01

Mean deviation: 0.01%.

3.2 Monte Carlo Significance

Of 100,000 Monte Carlo trials with 6 randomly placed frequencies in the visible range, zero produced a mean Farey deviation equal to or less than the observed 0.01%.

$$p < 0.00001 \text{ (0 / 100,000)}$$

The complementary test using uniform ratio draws on [1.0, 1.65] also yielded 0/100,000 successes. The alignment is not an artefact of Farey fraction density within the octave.

3.3 The Cyan Prediction

The Farey fraction 4/3 (rank 7, between 5/4 = green and 3/2 = blue) maps to:

$$f(4/3) = 454 \times 4/3 = 605 \text{ THz (495 nm)}$$

This frequency corresponds to cyan, the blue-green hue at approximately 490-500 nm. Cyan is universally absent as a basic colour term. Of 110 languages in the World Color Survey, zero possess a basic term for cyan that is distinct from both green and blue [2,3]. In Berlin and Kay's original 98-language survey, no language has cyan as a basic colour term [1]. Languages that make a green-blue distinction (stage V and beyond in the implicational hierarchy) jump directly from green (5/4) to blue (3/2) without an intermediate cyan category.

The Arnold tongue model provides a quantitative explanation. The $4/3$ tongue (width proportional to $1/7^2 = 0.020$) is flanked by the $5/4$ tongue (width proportional to $1/9^2 = 0.012$) on the low side and the $3/2$ tongue (width proportional to $1/5^2 = 0.040$) on the high side. The $3/2$ tongue is the widest in the visible octave (excluding the fundamental). Its dominance means that the $4/3$ region is effectively absorbed: perceptual locking to the strong $3/2$ (blue) attractor and the moderately strong $5/4$ (green) attractor leaves no independent basin of attraction for $4/3$ (cyan).

This is a strong, falsifiable prediction: cyan should emerge as a basic colour term only in contexts where the blue and green categories are already well-established AND the perceptual coupling strength epsilon is sufficiently high to resolve the $4/3$ tongue independently. Consistent with this, cyan does appear as a secondary colour term in technologically advanced societies with extensive artificial colour environments (e.g., display technology), but never as a basic term in the Berlin-Kay sense.

3.4 Implicational Hierarchy and Farey Rank

The Berlin-Kay implicational hierarchy for chromatic terms is:

red -> green/yellow -> blue -> [brown, purple, pink, orange, grey]

The Farey ranks of the corresponding fractions are:

Colour	Fraction	Rank (p+q)	Berlin-Kay Stage
Red	1/1	2	Stage II (first chromatic)
Green	5/4	9	Stage IIIa
Yellow	7/6	13	Stage IIIb
Blue	3/2	5	Stage IV
Orange	8/7	15	Stage VII
Violet	5/3	8	(late / composite)

The Farey rank alone does not perfectly predict acquisition order: blue (rank 5) is acquired after green (rank 9) and yellow (rank 13), despite being a simpler fraction. This discrepancy is resolved by the two-factor model described in Section 4.

3.5 Warm-Before-Cool Asymmetry

The Berlin-Kay hierarchy exhibits a robust warm-before-cool asymmetry: red is always named first among chromatic terms, and yellow/orange emerge before blue/violet in most languages. In the Farey framework, this maps directly to interval proximity:

- * Warm colours (red, orange, yellow) occupy ratios $1/1$ through $7/6$ -- the near-fundamental region of the octave, within the first 17% of the frequency range above f_0 .
- * Cool colours (green, blue, violet) occupy ratios $5/4$ through $5/3$ -- the upper portion of the octave.

In Arnold tongue theory, near-fundamental intervals (those closest to the 1:1 resonance) lock first because they experience the strongest coupling to the fundamental. The warm-before-cool asymmetry is thus a direct consequence of proximity to the $1/1$ tongue: near-fundamental harmonics resolve before remote ones.

4. Discussion

4.1 Two Orthogonal Factors: Consonance and Salience

The Farey rank alone does not fully predict the Berlin-Kay acquisition order. Blue (3/2, rank 5) should, by consonance rank alone, be acquired before green (5/4, rank 9) and yellow (7/6, rank 13). Its later acquisition suggests that a second factor modulates the mapping from consonance rank to perceptual salience.

We propose that colour category formation is governed by two orthogonal factors:

1. Consonance rank (domain-general): The simplicity of the Farey fraction determines the width of the Arnold tongue and thus the intrinsic stability of the category boundary. Lower rank = wider tongue = more robust category.

2. Photopic luminosity weighting (domain-specific): The sensitivity of the human visual system is not uniform across the visible spectrum. The photopic luminosity function $V(\lambda)$ peaks sharply at 555 nm (540 THz), near the yellow-green boundary, and falls off steeply toward both spectral extremes [23]. This weighting amplifies categories near the sensitivity peak (yellow, green) and attenuates categories in the spectral tails (red, violet, and importantly, blue).

The effective salience of a colour category is the product of these two factors:

$$S(p/q) = W(p/q) \times V(f_0 \times p/q)$$

where W is the Arnold tongue width and V is the photopic luminosity at the corresponding frequency. This two-factor model resolves the blue anomaly: blue (3/2) has wide Arnold tongue width but low photopic sensitivity ($V(475 \text{ nm}) = 0.09$, compared to $V(555 \text{ nm}) = 1.00$ for yellow-green), placing it below green and yellow in effective salience despite its superior consonance rank.

Table 3. Two-factor salience model.

Colour	p/q	Rank	$W \sim 1/(p+q)^2$	$V(\lambda)$	$S = W \times V$	Predicted Order	Observed Order
Red	1/1	2	0.250	0.004	0.0010	--	1
Yellow	7/6	13	0.006	0.995	0.0059	2	2-3
Green	5/4	9	0.012	0.954	0.0115	1*	2-3
Blue	3/2	5	0.040	0.091	0.0036	3	4
Orange	8/7	15	0.004	0.265	0.0011	5	5+
Violet	5/3	8	0.016	0.0004	0.000006	6	6+

(*Red's primacy requires a third factor: it sits at the fundamental itself, 1/1, which has categorically different status from any overtone. The fundamental is not merely the widest tongue but the reference point of the entire octave structure.)

The two-factor model correctly predicts: (i) red first (fundamental); (ii) green and yellow next (high luminosity x moderate-to-low rank); (iii) blue after green/yellow (high rank but low luminosity); (iv) orange and violet last (high rank and/or low luminosity).

4.2 Connection to Musical Consonance

The Farey fractions appearing in colour categorisation are identical to the intervals of musical consonance within the just intonation system:

Colour	Farey Fraction	Musical Interval
Red	1/1	Unison
Orange	8/7	Septimal second
Yellow	7/6	Septimal minor third

Green	5/4	Major third
Cyan (absent)	4/3	Perfect fourth
Blue	3/2	Perfect fifth
Violet	5/3	Major sixth

This is not a metaphorical correspondence. The mathematical structure is identical: the same set of simple rational fractions that divide one octave of acoustic frequency into perceptually distinct intervals also divides one octave of electromagnetic frequency into perceptually distinct colour categories. The mechanism -- Arnold tongue locking in a nonlinear perceptual system spanning an approximately 2:1 frequency range -- is domain-general. The specific realization differs (cochlear mechanics vs. photoreceptor absorption spectra), but the organising principle is the same.

The parallel extends to the "missing" interval: in music, the perfect fourth (4/3) has ambiguous consonance status -- it is considered consonant in some contexts and dissonant in others, depending on the harmonic series context [15]. Its colour analogue, cyan, exhibits precisely the same ambiguity: universally absent as a basic term, yet readily perceived and named in enriched colour environments.

4.3 Arnold Tongue Overlap and the Cyan Gap

The absence of cyan as a basic colour term can be modeled quantitatively using Arnold tongue overlap theory. When two tongues overlap in parameter space, the system cannot stably lock to both simultaneously; the wider tongue typically dominates and absorbs the narrower one [16,18].

At the 4/3 position in the visible octave (605 THz), the perceptual system is within the overlap region of three tongues:

- * The 5/4 tongue (green), extending upward from 567 THz
- * The 4/3 tongue (cyan), centered at 605 THz
- * The 3/2 tongue (blue), extending downward from 681 THz

The 3/2 tongue is by far the widest (rank 5), and the 5/4 tongue is moderately wide (rank 9). The 4/3 tongue (rank 7) sits between them but cannot establish an independent locking region because the combined extent of the flanking tongues covers the 4/3 position. The result is that the 605 THz region is partitioned between green and blue, with no independent cyan category.

This analysis predicts specific properties of the green-blue boundary in languages at different stages of the implicational hierarchy:

- * In stage I-II languages (no green/blue distinction), the 5/4 through 3/2 region is a single undifferentiated "grue" category.
- * In stage III-IV languages (green and blue distinguished), the boundary falls near 4/3 (605 THz, ~495 nm), which corresponds precisely to the observed green-blue boundary in the WCS data [2].
- * In no language does an independent cyan category emerge between green and blue as a basic term.

4.4 Why This Was Invisible

The Farey structure of colour categories has been latent in published data since 1969. The Berlin-Kay data contains the spectral locations. The Farey sequence has been studied since the 19th century. Arnold tongue theory has been applied to perception since the 1980s. The connection was not made because colour science and acoustics/dynamical systems theory occupy different disciplinary spaces.

The key conceptual step is recognising that the visible spectrum spans one octave -- a fact that, while occasionally noted, has not been treated as a structural constraint on colour perception. Once the octave framing is adopted, the Farey structure is immediately visible in the standard spectral data.

This is a general pattern: domain-specific fields accumulate precise measurements within their own coordinate systems but lack the cross-domain perspective to recognise when the same mathematical structure appears in different physical substrates. The Consonance Principle, which predicts Farey hierarchy locking wherever coupled oscillators span rational frequency ranges, provides the unifying framework.

4.5 Predictions and Falsifiability

The Farey model generates several testable predictions beyond those already confirmed:

1. Category boundary precision. The sharpness of colour category boundaries should correlate with Arnold tongue width: the red-orange boundary (between $1/1$ and $8/7$, both high-rank) should be less sharp than the green-blue boundary (between $5/4$ and $3/2$, one of which has very low rank). This is testable with the WCS naming consistency data.

2. Cross-linguistic boundary locations. The green-blue boundary should cluster near $4/3 \times f_0 = 605$ THz (~ 495 nm) across all languages that make the distinction, regardless of language family. Preliminary inspection of WCS data supports this [2,5].

3. Artificial spectral environments. In populations with extensive exposure to artificially enriched colour environments (display technology, printing), cyan should show incipient categorisation -- not as a basic term (which requires the Arnold tongue to separate from blue and green) but as a reliably named secondary category. This is observationally confirmed in industrialised populations.

4. Tetrachromat predictions. Human tetrachromats, who possess a fourth cone type and thus a wider effective spectral range, should show colour categories at Farey fractions of their extended octave. If the tetrachromat visible range extends to approximately 2 octaves, additional categories at fractions like $7/4$ and $9/5$ are predicted.

5. Non-human colour vision. Species with different visible spectral ranges should show colour discrimination peaks at Farey fractions of their specific octave. Mantis shrimp, with 12-16 spectral channels spanning approximately 300-720 nm (a range exceeding 2:1), provide a natural test case: their spectral sensitivity peaks should align with Farey fractions of their visible range.

4.6 Relation to Prior Work

Several researchers have noted partial correspondences between colour and music. Newton famously divided the visible spectrum into seven colours by analogy with the musical scale [12], but his mapping was arbitrary (he imposed a diatonic scale without physical justification) and has been dismissed as numerological. Our result differs fundamentally: we do not impose a musical analogy but rather show that the independently measured spectral locations of cross-linguistically universal colour categories fall at Farey fractions with $p < 0.00001$ significance. The structure was not assumed but discovered in existing data.

Pridmore [26] and Solli and Lenz [27] have explored wavelength-based periodicity in colour appearance but did not frame the analysis in terms of frequency ratios, Farey sequences, or Arnold

tongues, and did not connect to the Berlin-Kay universals. Our contribution is the identification of the specific mathematical framework (Farey hierarchy, Arnold tongue locking) that unifies colour categorisation with consonance phenomena across multiple physical domains.

5. Conclusion

The six universal chromatic colour categories of human language sit at exact Farey sequence fractions of the visible electromagnetic octave, with a mean deviation of 0.01% ($p < 0.00001$). The seventh low-rank fraction, $4/3$ (cyan), is universally absent as a basic colour term, as predicted by Arnold tongue overlap theory. The Berlin-Kay implicational hierarchy follows from a two-factor model combining Farey consonance rank (domain-general) with photopic luminosity weighting (domain-specific).

These findings reframe a 57-year-old puzzle in colour science. The question is no longer "why do languages partition colour the way they do?" but rather "why would they partition it any other way?" The visible spectrum spans one octave. Farey fractions are the unique, mathematically optimal partition of an octave into maximally distinct intervals. The human visual system, like any nonlinear oscillator driven across an octave of input frequency, locks to these fractions. The Berlin-Kay universals are not arbitrary conventions, biological accidents, or statistical reflections of the environment. They are the inevitable consequence of rational-fraction locking in a perceptual system that spans one octave of electromagnetic frequency.

The convergence of this result with independently documented Farey hierarchy locking in musical consonance [14,15], exoplanet orbital architectures [19,20], planetary ring structures [28], BCS superconductor gap ratios [29], and atomic emission spectra [30] suggests that consonance hierarchy is a universal organising principle that transcends physical substrate. Colour categories, musical intervals, and orbital resonances are different expressions of the same mathematical structure: the Farey sequence, made physical by Arnold tongue dynamics.

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Acknowledgments

This work was conducted in collaboration with Claude (Anthropic), whose cross-domain pattern recognition contributed to identifying the Farey structure in published colour data. The author acknowledges the World Color Survey research group for making their data publicly available, and the communities maintaining the NIST, CIE, and Munsell spectral databases.

Data Availability

All data used in this analysis are from published sources cited in the reference list. The World Color Survey data are available at <https://www1.icsi.berkeley.edu/wcs/>. CIE spectral data are available through the CIE website. Computation code is available from the author upon request.

Competing Interests

The author declares no competing interests.

Submitted for consideration to Nature Human Behaviour / PNAS

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